PRE-STARTUP MEETING

- Present typical GE performed startup tasks for heavy duty machines to Customer/Architect Engineer (AE)/Contractor.
- Customer/AE/Contractor presents site specific unit/plant startup tasks and schedule.
- GE and Customer/AE/Contractor thoroughly understand startup process and schedule.
- GE and Customer/AE/Contractor assign responsibilities for checkout, calibration, and testing, and resolve any responsibility conflicts. Key areas discussed include device calibration, generator protective relaying and metering, generator transformer and high yard, performance, reliability, and emissions testing requirements, obligations and procedures, Bently Nevada vibration systems, fire protection, auxiliaries (GE and customer supplied), and other "overlapping" systems.
- GE and Customer/AE/Contractor agree to planned startup schedule.
- Additional Items

  Customer/AE/Contractor will provide construction personnel with applicable drawings.

  GE provides Customer/AE/Contractor with list of equipment required for Mark IV and Excitation check-out. Focusing on long-lead items, sources and timing of equipment are obtained and agreed upon.

  GE provides instructions and Device Summary to Customer/AE/Contractor to expedite calibration of turbine devices. GE also informs Customer/AE/Contractor that all device calibration data is required for the startup report.

  Note: The above guidelines do not take into account GE supplied 'non-typical' options and/or auxiliaries.

ASSESS SITUATION AFTER ARRIVAL ON SITE

- Meet responsible parties.

  1. Project Manager/Startup Manager/Electrical Supervisor(s)
  2. Mechanical Supervisor(s)

- Learn of target startup date.

  Review project schedule and progress to date.

  Safety will NOT be compromised for scheduling considerations or any other reason.

  Discuss Tagout/Lockout procedures.

- Determine if all necessary drawings and documentation are on site. Includes the following:

  Cable Block Diagram, Interconnection Diagram, Speedtronic Elementary, Connection Diagram, Generator Control Panel and GAC Drawings, Control Specifications, Device Summary, Package Wiring Drawing, Motor Control Center, all vendor supplied Equipment Drawings, etc.

- Determine completeness of construction.

  Answer electrical construction questions (conduit routing, cable routing, cable termination, etc.).

  ALL field-connected shielded wire/cabling should have each drain wire individually insulated and terminated on the Analog I/O Module Shield Ground Bus Bar closest to the terminal board where the signal wires terminate.

  Check to see if all interconnecting wiring and cabling is pulled and terminated (GE supplied and customer supplied).

  Review mechanical construction of Gas Turbine and Gas Turbine Auxiliaries.

  Develop method and assign manpower to ensure all systems are per piping schematics.

  Verify installation of all Orifice Plates per appropriate piping schematics.

  Review completeness of customer supplied systems (i.e., fuel gas/oil, customer supplied piping, exhaust system, high voltage system, etc.).

  Review status of FRNs FMIs, TILs, SALs, if applicable.

- Determine unit complexity.

  Standard and Non-Standard Options – determine responsibility for checkout and development of testing procedures.

- Determine when permanent 480 VAC power will be available (and 125 VDC, if applicable) to MCC/PCC.

- Identify support personnel (i.e., technicians, electricians, etc.).

- Determine Tagging/Lockout Procedures to be followed on site during Startup Activity.

  Determine required levels of activity to meet schedule and communicate the following (as appropriate) to Customer/AE/Contractor and interested parties.

- GE activities.

- Contractor activities.

PRE-STARTUP ACTIVITIES

- Assemble and mark up necessary drawings.

- Gather Auxiliary Motor/Equipment nameplate data for Startup Report.

- Gather all other Nameplate Data required for Startup Report.

- Gather Calibration Data for Startup Report.

- Perform ground checks.

  Megger incoming power leads.

- Perform battery installation and checks, including the following:

  1. Electrolyte levels.

  2. Connections (i.e., tightness, cleanliness, conductive grease, etc.).

  3. Polarity.

  4. Voltages.

5. Vents.

- Identify Gas Turbine MCC/PCC distribution panel circuits.

  Mark up Distribution Panel Identification Sheets per Speedtronic Elementary/MCC/PCC drawings.

  Resolve discrepancies noted and document changes, if necessary.

- Open all Speedtronic Panel Power Supply switches.

- Remove all fuses, open all circuit breakers in Speedtronic Panel, MCC, PCC, Generator Control Panel, Generator Accessory Compartment, Auxiliary Control Panel, etc., and follow Site or GE Tagging/Lockout Procedures for personnel protection.

- Apply 480 VAC power to unit MCC/PCC.

  Verify phasing.

- Apply power to Generator Space Heaters.

  Megger circuits to verify continuity and ground impedance.

- Power-up lighting circuits.

- Power-up battery charger.

  Follow Manufacturer's instruction re: breaker closure sequence, equalizing charge (if applicable), etc.

  Monitor Battery condition during Startup Activities (i.e., electrolyte levels, charging current, etc.).

- Speedtronic Panel Power-up.

  Refer to Speedtronic Maintenance Manual for specific pre-checkout and power-up procedures.

  Verify Berg jumper placement and position; document changes if necessary.

  Check for grounds on load side of applicable fuses before inserting fuses.

  Power-up Processor Power Supplies (<C>), followed by (<RST>), individually).

Contact Input Modules (isolate and resolve ground(s) prior to inserting CIM fuses).

Relay Modules (isolate and resolve ground(s) prior to inserting fuses for <RELAY 1>).

Auxiliary Power Supplies, etc., (clear all shield grounds prior to powering-up <PS.REL1>, <PS.REL2>, and <P28X>).

- Resolve all Diagnostic Alarms as soon as possible.
- Begin Controls System Checkout by resolving all "erroneous" Alarm Drops.
- Monitor Ground Impedance during Startup Activities daily. Troubleshoot and resolve low readings (i.e., less than 500K ohms) in a timely manner.
- Make and compare <C> and <RST> Sequencing File Dumps or rung display when applicable against Speedtronic Elementary.

  Report all discrepancies to Installation Support/Controls Engineering for resolution.

  Note additions and deletions required to make As-Shipped software agree with Speedtronic Elementary.

  Make and document changes as necessary.
- Make and compare Control Constant printouts against Control Specifications - System Settings Drawing.

  Note errors/omissions/additions required to bring Control Constants Display into agreement with Control Specification and System Settings Drawing.

  Report all discrepancies to Installation Support/Controls Engineering for resolution.

  Make changes to Control Constant Display/Control Specification - System Settings Drawing, if necessary, and document.
- Loop-check Contact Inputs.

  Manipulate device contacts and incorporate device calibration (temperature switch, pressure switch, limit switch, control switch, etc.) if possible. If not, lift or jumper wires (as appropriate) at device wire termination in sub-junction box closest to device.

  Correct wiring errors and document changes if necessary.

Identify faulty components and repair or replace.

Determine if correct contacts (i.e., N.O. or N.C.) are depicted on factory drawings; document any changes required for proper operation/proper depiction on Speedtronic Elementary and/or Device Summary.

- Loop-check Solenoid Circuits.

  Energize/de-energize solenoids via Logic Forcing; observe or listen for actuation, as appropriate.

  Correct wiring errors and document changes if necessary.

  Identify faulty components and repair or replace.

  Using an Auxiliary Air Supply whenever possible, verify proper operation of all Pneumatic Valves and/or devices.

- Loop-check Thermocouple/RTD Circuits.

  Lift T/C or RTD wire at junction closest to sensor (i.e., not interconnecting or extension wire).

  Correct wiring errors and document changes if necessary.

  Identify faulty components and repair or replace.

- Loop-check Transducer Inputs (should be done in conjunction with Control Panel calibration).

  Verify calibration range (using the Control Specification and checking Device Summary, Speedtronic Elementary, or Piping Arrangement Parts List) and Control Constant scaling values. Document changes if necessary.

  Adjust Transducer "zero" and "span" potentiometers for correct voltage or milli AMP output, then use signal to scale appropriately on Speedtronic display.

  Correct wiring errors and document changes if necessary.

  Identify faulty components and repair or replace.

- Loop-check Relay Outputs.

  Energize/de-energize relays via Logic Forcing; observe actuation of connected device (contactor, indicating lamp, permissive logic, etc.).

  Correct wiring errors and document changes, if necessary.

  Identify faulty components and repair or replace.

- Loop-check MCC/PCC Circuits.

  Remove TOL heater elements prior to initially powering-up MCC bucket and during checkout, as appropriate.

  Correct wiring errors and document changes if necessary.

  Identify faulty components and repair or replace.

- Motor Power Circuit Checkout.

  Megger power circuits; record readings for Startup Report.

  Bump check motors for proper rotation; correct phasing/polarity if necessary.

- Assist With Performance of L.O./Hydraulic Flush.

  Controls interface (i.e., ensure ALL flushing blocks installed on all servos, Aux. Motor Starter operation, etc.).

  Check Hydraulic Pump Motor current at start of flush and adjust volume to prevent over-current, if necessary.

- Speedtronic Algorithm Checkout and Calibration; Refer to Control Specification (System Adjustments).

  - Electrical Overspeed Checks.

    HP Shaft.

    LP Shaft, if applicable.

    IP Shaft, if applicable.

- Speed Level Circuit Checks.

  HP Shaft.

  LP Shaft, if applicable.

  IP Shaft, if applicable.

- Temporarily disconnect and resistance check all Magnetic Speed Pickups at Control Panel Check Pickup Gap, if possible.

- Over-temperature protection algorithm checkout.

- Primary temperature control bias algorithm checkout.

- Backup temperature control bias algorithm checkout.

- L.O. temperature monitor algorithm checkout, if applicable.

- Wheelspace temperature monitor algorithm checkout, if applicable.

- Fuel Valve Servo Loop-checks (polarity/control).

  Stop-Speed Ratio Valve.

  Gas Control Valve.

  Liquid Fuel Bypass Valve.

- Fuel Valve LVDT Calibrations.

  Stop-Speed/Ratio Valve (observe Berg jumper placement for calibration - see Control Spec. - Sys. Settings; return Bergs to running configuration when calibration is complete).

  Gas Control Valve.

  Liquid Fuel Bypass Valve (observe Berg jumper placement for calibration - see Control Specification).

  System Settings (return Bergs to running configuration when calibration is complete).

- Vibration System Checkout, if applicable.

- Miscellaneous Servo Loop checks (polarity/control), if applicable.

  Inlet guide vanes, second-stage nozzles, etc.

- Miscellaneous LVDT calibrations, if applicable.

  Inlet guide vanes, second-stage nozzles, etc.

- Wet low NOx algorithm checkout, if applicable.

- Plot reference curves.

  Plot protective limit curves.

  Verify sequencing logic/timers.

- Miscellaneous algorithm checkout(s), if applicable.

  Document changes if necessary.

- Station Plant Controls Interface Checkout, if applicable.

  Correct wiring errors and document changes if necessary.

  Assist with sequencing/logic questions; incorporate and document changes, if necessary.

WARNING: If hydrogen cooled generator, follow all recommended procedures for testing and filling hydrogen cooled generators.

- Generator Controls Interface Checkout, if applicable, and Excitation Checkout.

- Load compressor controls and interface checkout, if applicable.

WARNING: Ensure fire protection systems (especially CO2) are disabled until required for service.

- Fire detection/protection system checkout.

  Remove discharge nozzles, and air-blow extinguishing agent piping; replace discharge nozzles and restore piping.

  Loop check detectors/annunciator.

  Functionally check discharge means.

Assist with "puff-test"/concentration test, as required, to comply with local regulation and/or insurance requirements.

If Chemetronics Micro 1 EV Control Panel, program all options and timers.

- Ignition Circuit Checkout.

  Remove ignitor assembly, ground, and energize Ignition Transformer(s)/Ignitor(s) via Logic Forcing.

- Air Inlet Filter Self-Sequencer Checkout, if applicable.

  Initiate via Pressure Switch, then Manual Pushbutton.

- Blow Air Inlet Filter HP Piping to remove debris/slag, etc.

- Flame Detector Checkout.

WARNING 300 VDC To Honeywell Flame Detectors - Use Caution When Testing.

- Blow Fuel Gas Piping to remove debris/slag, etc., if applicable.

- Flush Liquid Fuel Piping to remove debris/slag, etc., if applicable.

- Blow Steam Injection Piping, flush Water Injection Piping.

- Flush Evaporative Cooler Water Piping to remove debris/slag, etc., if applicable.

- Verify cleanliness of Inlet Ductwork.

- Verify cleanliness of Exhaust Ductwork.

- Verify Valve Line-up of All Piping Systems, including:

  Cooling and Sealing Air.

  Cooling Water.

  Accessory Gage Cabinet.

  Seal Oil System, if applicable.

- Prior to shaft rotation, DC lube oil pump control circuit, speed, and output pressure MUST be checked.

- Prior to Startup, review alignment, calibration, and checkout data. DO NOT PROCEED if As-Found and As-Left data is not complete for critical devices.

STARTUP ACTIVITIES

- Prior to Startup, ensure that fuel, steam, and water supply conditions are correct per Control Specification and/or piping schematics. Prior to operation, also ensure that fuel, steam, water, and lube oil meet Design Specifications.

CAUTION: Observe all equipment limitations as outlined by Instruction Manuals - Operating sections.

- Refer to "Checks with Unit Running" Section of Control Specification and complete all recommended tests.

- Refer to Instruction Manuals - Operating sections for other applicable checks.

- Initial Crank checks, Slow Roll Hydraulic Ratchet.

  Starting Circuit Checkout.

  Listen for rubs.

- Initial Firing.

  Flame Detector circuit checkout.

  Monitor vibration levels, exhaust temperatures, fuel flow, etc.

  Check for fuel leaks.

- Mechanical Overspeed Checks.

  HP Shaft.

  LP Shaft, if applicable.

- Excitation/Synchronization Checks (assuming potential source excitation system, if generator application) - Shaft Ground Brushes mounted?

  Field Flashing.

  Excitation checks (AC and DC regulators).

Regulator Transfer.

Phasing (across High-Voltage Generator Breaker stabs).

Manual Synch. circuit checks (with Gen. Brkr. in "Test" position).

Automatic Synch. circuit checks (with Gen. Brkr. in "Test" position).

- Initial Synchronization and Loading, if Generator Application.

  Low Power operation for:

  1. Data gathering.
  2. Protective Relay and CT phasing checks/verification.

- Secondary Fuel Startup Checkout.

OPERATIONAL SUPPORT AND TROUBLESHOOTING

- Temperature Control.
- Load Control.
- Wet Low NOx.
- Performance and Emissions Testing, if necessary.
- Fine Tune Unit Operation.
- Startup Remaining Unit Auxiliaries (e.g., Water Wash, Compressor Cleaning).
- Gather Operating Data for Startup Report.

DRAWING AND DOCUMENTATION

- Complete Startup Report and return to Installation Support.
- Mark up Drawings listed below and return to Installation Support:

  Factory As-Builts.

  GE District Office As-Builts.

  Intermediate Customer As-Builts.

POST-INSTALLATION MEETING

- Ensure all GE startup tasks are complete per Pre-installation Meeting agreements.

DEMOBILIZATION

- Ensure all documentation, computer equipment, and test equipment used for Startup is removed from site.